

ULTRASONIC THICKNESS GAUGING & NDT INSPECTION REPORT

Report Reference:	INSP-2025-084-UT	Inspection Date:	18 September 2025
Plant Asset Tag:	CDU-Pipe-104	Plant Unit:	Crude Distillation Unit (CDU-II)
Governing Code:	ASME B31.3 / API 570	NDT Method:	Ultrasonic Pulse-Echo (UT)
Pipe Material:	ASTM A106 Grade B Seamless	Design Pressure:	3.2 MPa (464.1 psig)
Nominal Thickness:	12.7 mm (0.500 in)	Measured Thickness:	7.2 mm (0.283 in)
Corrosion Rate:	0.45 mm / year	Design Temperature:	180 deg C (356 deg F)

1. ULTRASONIC SCANNING & NDT ASSESSMENT FINDINGS

Ultrasonic thickness measurements were acquired across 16 grid locations along the pre-heat crude transfer header (CDU-Pipe-104) utilizing calibrated dual-element 5.0 MHz transducers. The nominal design wall thickness is 12.7 mm. The localized minimum measured wall thickness recorded is 7.2 mm along the 6 o'clock invert, representing a cumulative wall loss of 43.3% due to high-temperature naphthenic acid and sulfidic erosion-corrosion. Current localized corrosion rate is established at 0.45 mm/year.

2. ASME B31.3 CHAPTER II STATUTORY CODE COMPLIANCE

Under ASME B31.3 Paragraph 304.1.2, minimum required pressure design wall thickness t_d is 3.14 mm for internal design pressure 3.2 MPa. Factoring in statutory retirement allowance, the minimum allowable retirement thickness is 5.12 mm. Since the current measured wall thickness (7.2 mm) exceeds t_{min} (5.12 mm), the transfer line retains +2.08 mm of usable corrosion margin, corresponding to an estimated 4.62 years of safe operation prior to reaching retirement threshold.

STATUTORY APPROVAL & RE-COMMISSIONING DIRECTIVE:

- Fitness-for-Service (FFS): LEVEL 1 CERTIFIED per API 579-1 / ASME FFS-1
- Action Required: Plant Superintendent digital sign-off and 12-month UT re-inspection protocol.

Lead NDT Inspector: R. K. Sharma (Level III UT/RT #98421)

Plant Superintendent Approval: PENDING AUTHORIZATION